

[ENTER COMPANY NAME]

QUALITY MANAGEMENT SYSTEM

STANDARD OPERATING PROCEDURE

Design Controls and Technical File Architecture

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Applicable Regulatory Standards	ISO 13485:2016 Clause 7.3 FDA 21 CFR Part 820.30 (Design Controls) EU MDR 2017/745 Annex II & III

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1.0 Purpose

This Standard Operating Procedure (SOP) defines the mandatory controls, phase-gated reviews, and compilation requirements for the design and development of medical devices. The primary objective is to verify that development projects systematically translate user needs into fully validated physical devices, ensuring safety, efficacy, and regulatory compliance before commercial transfer.

2.0 Scope

This procedure is strictly applicable to all product design activities, core platform modifications, user-interface alterations, and subsequent technical line extensions developed by the organization. The process controls govern the entire lifecycle spanning from formal project initiation through to manufacturing transfer and final technical file compilation.

3.0 Responsibility and Authority

3.1 Program Management / Project Lead

Responsible for formulating the multi-disciplinary project team, establishing the milestone design plan, maintaining the project schedule, and compiling the active Design History File (DHF).

3.2 R&D / Design Engineering Team

Responsible for executing technical design configurations, generating physical layouts, modeling blueprints, populating technical engineering specifications, and performing physical prototyping.

3.3 Independent Reviewer

A qualified technical professional who is not directly involved in the specific design phase under evaluation. Responsible for providing impartial technical oversight during formal design review stages.

4.0 Procedure

4.1 Phase-Gated Design Matrix

To ensure robust control, all development initiatives must advance sequentially through five distinct, documented design gates. Progression across a gate demands a formal design review session and quality authorization.

Design Phase	Core Deliverables & Procedural Mapping	Required DHF Documentation Artifacts
Phase 1	Design Planning & Input Formulation: Identify intended use profiles, user needs, and primary human factors. Establish regulatory strategy pathways.	Design and Development Plan, Product User Needs Specification, Regulatory Strategy File.
Phase 2	Design Outputs & Prototyping: Translate technical inputs into clear physical outputs. Create engineering models, software source trees, and schematic layouts.	Product Engineering Specifications, Risk Analysis Records (DFMEA), Mechanical CAD Models, Bill of Materials (BOM).
Phase 3	Design Verification: Prove that the physical design outputs match the design inputs. Execute mechanical, electrical, and biocompatibility lab testing.	Verification Protocols, Certified External Lab Test Reports, Traceability Matrix (Inputs to Outputs).
Phase 4	Design Validation: Prove that the device consistently meets user needs and intended clinical uses in simulated or actual clinical environments.	Validation Protocols, Human Factors Evaluation Reports, Clinical Evaluation Reports (CER).
Phase 5	Design Transfer: Transfer the validated design into commercial production layouts. Validate cleanroom tooling and final inspection steps.	Device Master Record (DMR) Index, Process Validation Protocols (IQ/OQ/PQ Reports), Design Transfer Sign-off.

4.2 Design Traceability Matrix (DTM) Mandate

The project team must maintain a comprehensive Design Traceability Matrix (DTM) throughout the project lifecycle. The matrix must map each User Need to its corresponding Design Input, link each Input to its corresponding Design Output, and cross-reference each Output to its specific Design Verification and Validation protocols. A project cannot clear Phase 4 review if any trace pathway remains unverified or disconnected.

REGULATORY TRACEABILITY MAPPING: The traceability matrix is the primary structural asset evaluated during Notified Body and FDA QMS inspections. It provides objective evidence that all product hazards identified in the Risk Management File have been systematically addressed by design mitigations or verified by physical testing.

4.3 Design Review Sessions

Formal design reviews must be conducted at the conclusion of each phase gate. Review panels must include the project originators, department managers, quality engineers, and at least one Independent Reviewer. All meetings must be documented in formal minutes, detailing any action items, technical challenges identified, and the final pass/fail authorization status.

4.4 Tech File Architecture (EU MDR Alignment)

Parallel to compiling the DHF, the Regulatory Affairs Specialist is responsible for assembling the product Technical Documentation for global regulatory submissions. This file must be structured according to STED (Summary Technical Documentation) formats and must systematically include:

- Detailed device description, variant breakdowns, and product specifications.
- Comprehensive labeling assets, packaging drawings, and active Instructions for Use (IFU).
- GSPR (General Safety and Performance Requirements) Checklist, detailing the specific methods used to demonstrate compliance with EU MDR Annex I.
- Pre-clinical and clinical data validation sets (Biocompatibility, Sterilization validation, stability summaries).

5.0 Documentation Retention and Archiving

The Design History File (DHF) and formal Technical Documentation files are critical quality assets. They must be maintained in secure, fire-walled electronic repositories for a minimum of ten (10) years for standard devices, and fifteen (15) years for implantable technologies, counted from the date the last unit of the device family is formally distributed to commercial markets.

6.0 Associated Reference Materials

- **SOP-QA-001:** Document and Data Control System
- **SOP-QA-002:** Change Control System
- **SOP-QA-007:** Risk Management Program (ISO 14971 Integration)
- **Form-RD-005A:** Design Traceability Matrix (DTM) Template

7.0 Revision History Log

Revision Level	Effective Date	Author / Originator	Detailed Description of Modifications
00	[Date]	[Name], R&D Lead	Initial baseline release of formalized Design Controls and Technical File Architecture Standard Operating Procedure.